

DevOps Series

Deploying Graylog Using Ansible

This 11th article in the DevOps series is a tutorial on installing Graylog software using Ansible.



Graylog is a free and open source log management software that allows you to store and analyse all your logs from a central location. It requires MongoDB (a document-oriented, NoSQL database) to store meta information and configuration information. The actual log messages are stored in Elasticsearch. It is written using the Java programming language and released under the GNU General Public License (GPL) v3.0.

Access control management is built into the software, and you can create roles and user accounts with different permissions. If you already have an LDAP server, its user accounts can be used with the Graylog software. It also provides a REST API, which allows you to fetch data to build your own dashboards. You can create alerts to take actions based on the log messages, and also forward the log data to other output streams. In this article, we will install the Graylog software and its dependencies using Ansible.

GNU/Linux

An Ubuntu 16.04.3 LTS guest virtual machine (VM) instance will be used to set up Graylog using KVM/QEMU. The host system is a Parabola GNU/Linux-libre x86_64 system. Ansible is installed on the host system using the distribution package manager. The version of Ansible used is:

```
$ ansible --version
ansible 2.4.1.0
  config file = /etc/ansible/ansible.cfg
  configured module search path = [u'/home/shakthi/.ansible/
```

```
plugins/modules', u'/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python2.7/site-
packages/ansible
  executable location = /usr/bin/ansible
  python version = 2.7.14 (default, Sep 20 2017, 01:25:59)
  [GCC 7.2.0]
```

Add an entry to the `/etc/hosts` file for the guest 'ubuntu' VM as indicated below:

```
192.168.122.25 ubuntu
```

On the host system, let's create a project directory structure to store the Ansible playbooks:

```
ansible/inventory/kvm/
  /playbooks/configuration/
  /playbooks/admin/
```

An 'inventory' file is created inside the `inventory/kvm` folder that contains the following code:

```
ubuntu ansible_host=192.168.122.25 ansible_connection=ssh
ansible_user=ubuntu ansible_password=password
```

You should be able to issue commands using Ansible to the guest OS. For example:

```
$ ansible -i inventory/kvm/inventory ubuntu -m ping
```